

Pension System Reform

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1. Introduction

TBD.

Related Literature:

- [Bi and Zubairy \(2023\)](#). Empirical paper on the mechanisms of pension reforms. Three mechanisms are important: income effect on working longer caused by benefit cut, early withdrawal to extract more benefits, and the uncertainty mechanism due to implementation uncertainty.

2. Model

Time is continuous.

2.1. Population

Let $b(s)$, $m(s)$ be the fertility rate and the mortality rate respectively. Let $g(t, s)$ be the unnormalized population density. The dynamic is given by

$$\frac{\partial g}{\partial t} = -\frac{\partial g}{\partial s} - mg + \int b g ds \delta_0. \quad (1)$$

2.2. Pension System

Households retire at age T . When retiring, the households can choose between a lumpsum transfer ℓ_t and a pension that pays stream of income p_t . Let $Q(t, s)$ be the population density among those on pension. The dynamic is given by

$$\frac{\partial Q}{\partial t} = -Q_s - mQ + q(t)g(t, T)\delta_T \quad (2)$$

where q is the proportion of new retirees choosing pension. The government budget constraint is given by

$$dB_t = \left[rB_t + \tau y \int_0^T g(t, s) ds - (1 - q(t)) \ell_t g(t, T) - p_t \int_T^\infty Q(t, s) ds \right] dt. \quad (3)$$

2.3. Households Problem

The households choose between the two schemes to maximize the present value. We assume the type one extreme value disturbances on the two options. Hence

$$q_t = \frac{1}{1 + \exp\left(\ell_t - \mathbb{E}_t \left[\int_T^\infty e^{-(r(s-T) + \int_T^s m(x) dx)} p_s ds \right]\right)} \quad (4)$$

2.4. Pension Reform

Option 1: No Ponzi Scheme Condition

The government is forced to reform when it reaches the borrowing limit

$$\bar{B}_t = \int_t^\infty e^{-r(x-t)} \tau y \int_0^T g(x, s) ds dx = \int_0^\infty \int_0^T e^{-rx} \tau y g(x + t, s) ds dx. \quad (5)$$

When reaching the limit, the benefits are cut to zero.

Option 2: Zero Pension Fund

An alternative is to assume that the system collapse when the remaining fund reduces to $\bar{B}_t = 0$. This means that the “reform” is to cut the benefit and the pension tax simultaneously.

2.5. Equilibrium

Given initial condition B_0 , $Q_0 = 0$, and g_0 , the equilibrium is characterized by q_t, p_t such that

- (a) Given $p_s, s \geq t$, the optimal choice of the pension schemes aggregates to q_t .
- (b) q_t induces paths B_t, Q_t .
- (c) B_t, Q_t imply a path of p_t that is consistent with the one in (a).

We shall first deal with the population equation.

$$g(t, s) = \begin{cases} g(0, s - t) \exp\left(-\int_{s-t}^s m(x) dx\right), & s > t, \\ g(t - s, 0) \exp\left(-\int_0^s m(x) dx\right) & s \leq t. \end{cases} \quad (6)$$

The boundary condition is given by

$$\begin{aligned}
g(t, 0) &= \int_0^\infty b(s)g(t, s)ds \\
&= \int_0^t b(s)g(t-s, 0) \exp\left(-\int_0^s m(x)dx\right)ds + \int_t^\infty b(s)g(0, s-t) \exp\left(-\int_{s-t}^s m(x)dx\right)ds \\
&= \int_0^t b(t-u)g(u, 0) \exp\left(-\int_0^{t-u} m(x)dx\right)du + \int_t^\infty b(u)g(0, u-t) \exp\left(-\int_{u-t}^u m(x)dx\right)du
\end{aligned} \tag{7}$$

Let $S(s) = \exp\left(-\int_0^s m(x)dx\right)$ and $f(t) = g(t, 0)$. Assume that there is an age limit $\bar{s}$. The above equation can be rewritten as

$$f(t) = \int_0^t b(t-u)S(t-u)f(u)du + \mathbf{1}_{\{t \leq \bar{s}\}} \int_t^{\bar{s}} b(u)g(0, u-t) \frac{S(u)}{S(u-t)} du. \tag{8}$$

3. A Simplified Case (No Population Consideration)

Let N_t be the volume of population. bN_t is the inflow of newly born agents. Assume that the mortality rate is 0 before T , the age of retirement, and constant m after T .

Assume that the population age density is in the steady state. Let $\tilde{g}(t, s)$ denote the unnormalized density. We shall find the stable normalized density. Conjecture that

$$\tilde{g}(t, s) = g(s)N_t. \tag{9}$$

Plugging back to the population equation gives

$$g(s) \frac{\dot{N}_t}{N_t} = -\frac{\partial g}{\partial s} - m(s)g(s) + b\delta_0. \tag{10}$$

Notice that the right hand side does not depend on t . Hence $\frac{\dot{N}_t}{N_t} = n$ is a constant. We can also write the Dirac delta as part of the boundary condition:

$$g(s)n = -\frac{\partial g}{\partial s} - m(s)g(s), \quad g(0) = b. \tag{11}$$

This is a standard initial value problem with $m(s) = m$ for $s \geq T$ and 0 otherwise. The solution is given by

$$g(s) = \begin{cases} be^{-ns}, & s < T \\ be^{-ns}e^{-m(s-T)} & s \geq T. \end{cases} \tag{12}$$

Finally, the population growth n can be determined by the normalizing condition $\int_0^\infty g(s)ds = 1$:

$$1 = b \frac{1 - e^{-nT}}{n} + be^{-nT} \frac{1}{n+m}. \tag{13}$$

Next, we shall consider the pension system. The advantage of doing the simplification emerges in this part. Notice that in the current case, there is no need to keep track of the population density at every given ages. With an abuse of the notation, we shall denote the amount of people on pension by Q . The Q follows

$$\dot{Q}_t = q_t b e^{-nT} N_t - m Q_t. \quad (14)$$

Next, we can write down the government budget constraint.

$$\dot{B}_t = r B_t + N_t \frac{b \tau y}{n} (1 - e^{-nT}) - (1 - q_t) b e^{-nT} N_t \ell_t - p_t Q_t. \quad (15)$$

The borrowing limit is given by

$$\bar{B}_t = \int_t^\infty e^{-r(s-t)} N_s \frac{b \tau y}{n} (1 - e^{-nT}) ds = \frac{b \tau y}{n} (1 - e^{-nT}) \frac{N_t}{r - n}. \quad (16)$$

To determine the equilibrium, define the remaining time to reform

$$T^R(B_0, Q_0, N_0) = \inf \{s \geq 0 \mid B_s = \bar{B}_s\}. \quad (17)$$

Since the timing of reform is unique across cohorts, the remaining time to reform counts down to 0 deterministically at speed -1 . From this, using the definition of q_t ,

$$q_t = \frac{1}{1 + \exp\left(\ell - \frac{p}{r+m} (1 - e^{-(r+m)(T^R-t)})\right)} \quad (18)$$

for $t < T^R$. Now, we can solve for Q on $[0, T^R]$ and hence B as well.

To determine T^R , use the definition $\bar{B}_{T^R} = B_{T^R}$ and we have pinned down the equilibrium once T^R is determined.

4. Return on Pension Fund

In this section, we relax the assumption from the last section where the reform timing is certain by considering volatile investment return on the pension fund running by the government. Specificially, we shall assume that the fund follows a geometric Brownian motion. The new budget constraint is

$$dB_t = \left[r B_t + N_t \frac{b \tau y}{n} (1 - e^{-nT}) - (1 - q_t) b e^{-nT} N_t \ell_t - p_t Q_t \right] dt + \sigma B_t dZ_t \quad (19)$$

where Z_t is a Brownian motion. Now the reform time T^R becomes random. Next, we shall consider the detrend $\hat{B}_t = \frac{B_t}{N_t}$ and $\hat{Q}_t = \frac{Q_t}{N_t}$. The corresponding dynamics are given by

$$d\hat{B}_t = \left[(r-n)\hat{B}_t + \frac{b\tau y}{n}(1-e^{-nT}) - (1-q_t)be^{-nT}\ell - p\hat{Q}_t \right] dt + \sigma\hat{B}_t dZ_t \quad (20)$$

$$d\hat{Q}_t = [q_t be^{-nT} - (m+n)\hat{Q}_t] dt. \quad (21)$$

Note that the reform time can also be defined as the hitting time to $\hat{B}_t = 0$. The aggregate choice q_t now depends on $V(\hat{B}_t, \hat{Q}_t)$ where V is defined by

$$V(\hat{B}_t, \hat{Q}_t) = \mathbb{E} \left[\int_t^{T^R} e^{-(r+m)(s-t)} ds \mid \hat{B}_t, \hat{Q}_t \right]. \quad (22)$$

To solve for V , the corresponding PDE is given by

$$(r+m)V = 1 + \mu_{\hat{B}} \frac{\partial V}{\partial \hat{B}} + \mu_{\hat{Q}} \frac{\partial V}{\partial \hat{Q}} + \frac{1}{2} \sigma^2 \hat{B}^2. \quad (23)$$

Observe that $\hat{Q}$ stays in $[0, \frac{be^{-nT}}{m+n}]$. In the $\hat{B}$ coordinate, we use the following boundary conditions:

$$V(0, \hat{Q}) = 0 \quad \frac{\partial V}{\partial \hat{B}}(\hat{B}_{upper}, \hat{Q}) = 0.0. \quad (24)$$

The choice for the upper boundary comes from the fact that V is clearly increasing in $\hat{B}$ and is bounded above. It follows that the derivative of V must be small for sufficiently large $\hat{B}$.

5. Estimation

To estimate the unknown parameters, we use the following approximated maximum likelihood. Let Δ be the length of time gaps between observations and denote $X_t = (\hat{B}_t, \hat{Q}_t)'$. Let μ and Σ collect the drift term and the diffusion term respectively. The main challenge of maximum likelihood comes from computing $p(\Delta, X_{t+\Delta}, X_t)$, the transition probability from X_t to $X_{t+\Delta}$ after Δ . The following two approaches may be used.

Kolmogorov Forward Equation

Given θ , we can solve for the value function given by 22. Then we can solve for the Kolmogorov forward equation associated with the stochastic differential equation.

The drawback of this approach is the computational intensivity. To achieve the required accuracy, the grid size needs to be small enough. A local h -refinement algorithm needs to be used.

Local Gaussian Approximation with Method of Moments

This approach is motivated from the following Euler-Maruyama scheme. For small Δ , we may use the following approximation:

$$X_{t+\Delta} \approx X_t + \mu(X_t)\Delta + \Sigma(X_t)\sqrt{\Delta}Z \quad (25)$$

where $Z \sim N(0, I)$. Use the moments

$$B_{t+\Delta} - B_t \quad (26)$$

References

- Bi, H., & Zubairy, S. (2023, November). Public pension reforms and retirement decisions: Narrative evidence and aggregate implications. *American Economic Journal: Economic Policy*, 15(4), 142–82.